

1. What is Information Security? Explain the goals of Information Security and discuss how the CIA Triad helps protect organizational information.
2. Differentiate between Authentication, Authorization, and Accounting (AAA) with suitable examples.
3. An e-commerce company wants to ensure that online purchase agreements cannot be altered and that customers cannot deny submitting an order.
 - Explain how Digital Signatures provide security.
 - Describe how Integrity and Non-repudiation are achieved.
 - Recommend two additional security controls for the company.
4. Compare the Bell-LaPadula and Biba security models in terms of:
 - Objective
 - Security rules
 - Practical applications
5. A cloud storage service experienced unauthorized access due to weak passwords.
 - Identify two possible security threats.
 - Recommend suitable authentication or access control mechanisms.
 - Explain how security awareness training can reduce such incidents.
6. A company was infected with ransomware after an employee downloaded a malicious email attachment.
 - What immediate actions should the security team take?
 - Suggest long-term preventive measures.
 - Which preventive measure is the most effective? Justify your answer.
7. Define the following with one example each:
 - Confidentiality
 - Integrity
 - Availability
 - Threat
 - Vulnerability
8. A healthcare organization accidentally exposed patients' personal information through a misconfigured cloud server.
 - Which CIA principle(s) were violated?
 - Identify two vulnerabilities.
 - Recommend two security controls.

9. A company plans to implement Multi-Factor Authentication (MFA).

- Explain three advantages of MFA.
- Mention two implementation challenges.
- Recommend one additional authentication mechanism.

10. What is Access Control? Explain DAC, MAC, and RBAC with suitable examples.

11. A university uses a web portal for course registration. An attacker steals a student's login credentials.

- Which security principles have been violated?
- Recommend two authentication mechanisms.
- Explain why your recommendations improve security.

12. A financial institution is launching an online banking system.

- Identify major security risks.
- Recommend suitable countermeasures.
- Which countermeasure should receive the highest priority? Justify your answer.

13. What is Risk Management? Explain the following stages:

- Asset Identification
- Risk Identification
- Risk Evaluation
- Risk Treatment

14. A government office stores citizens' records in a centralized database.

- Identify important information assets.
- Evaluate possible security risks.
- Recommend suitable risk treatment options.

15. After conducting a risk assessment, an organization identified phishing, insider threats, and hardware failures.

- Prioritize these risks.
- Recommend an appropriate treatment strategy (avoid, mitigate, transfer, or accept) for each.

16. Explain the architecture of HDFS. Describe the functions of the NameNode, DataNode, and Secondary NameNode. Explain how MapReduce processes large datasets.

17. What is a Statistical Database? Explain:

- Data Compromise
- Inference Attacks (Snooping)
- Query Set Size Control
- One additional inference control technique

18. A research institute wants to protect ownership of its shared relational database.

- Why is database watermarking preferred over steganography?
- Explain how relational watermarking works.
- Mention two challenges of implementing database watermarking.

19. What is an Information Security Management System (ISMS)? Explain its objectives and discuss how it helps organizations manage information security risks.

20. A company lacks a formal Information Security Governance framework.

- Identify governance-related problems.
- Explain the role of Information Security Governance.
- Recommend governance practices that improve organizational security.

21. A manufacturing company is planning to implement ISO/IEC 27001-based ISMS.

- Discuss the benefits of implementation.
- Explain the challenges involved.
- Recommend whether the organization should adopt ISMS and justify your answer.